

Hdoc- Day 22

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Batch- 15

1. Algorithm

1. Start.
2. Declare integer variables `m`, `n`, `choice`, `i`, and `j`.
3. Accept values of `m` and `n`.
4. Check whether:
 - `m > 0`
 - `n > 0`
 - `m <= n`
5. If the input is invalid, display an error message and ask for `m` and `n` again.
6. Display the menu:
 - Multiplication tables of `m` and `n`
 - Multiplication grid
 - Consecutive integers
 - Exit
7. Accept `choice`.
8. Use a `switch` statement:
 - **Case 1:** Use a loop from 1 to 10 and display the multiplication tables of `m` and `n` side by side.
 - **Case 2:** Use nested loops. The outer loop runs from 1 to `m`, and the inner loop runs from 1 to `n`. Display `i * j` to form the multiplication grid.
 - **Case 3:** For every integer `i` from `m` to `n`, use an inner loop from 0 to 5 and display `i + j`. This prints the integer itself followed by the next five consecutive integers.
 - **Case 4:** Display an exit message and terminate the menu loop.
 - **Default:** Display an error message for an invalid menu choice.
9. Repeat steps 6–8 until the user selects option 4.
10. Stop.

2. Test-Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1 – Valid	<code>m=3, n=5, Choice 1</code>	Tables of 3 and 5 from $\times 1$ to $\times 10$ displayed side by side	Tables displayed correctly	Pass
2 – Valid	<code>m=3, n=5, Choice 2</code>	3×5 multiplication grid	Correct 3×5 grid displayed	Pass

3 – Valid	m=3, n=5, Choice 3	3: 3 4 5 6 7 8, 4: 4 5 6 7 8 9, 5: 5 6 7 8 9 10	Correct consecutive integers displayed	Pa ss
4 – Boundary	m=1, n=1, Choice 1	Table of 1 displayed in both columns	Correct output	Pa ss
5 – Boundary	m=1, n=1, Choice 2	Grid containing only 1	Correct output	Pa ss
6 –	m=0, n=5	Error because m is not positive	Error message	Pa
7 –	m=5, n=3	Error because m > n	Error message	Pa
8 – Invalid	m=3, n=5, Choice 9	Invalid menu choice message	Error message displayed	Pa ss
9 – Exit	m=3, n=5, Choice 4	Program exits	Exiting program.	Pa ss

3. C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int m, n, choice, i, j;
```

```
    /* Input and validation of m and n */
```

```
    do
```

```
    {
```

```
        printf("Enter two positive integers m and n (m <= n):  
");
```

```
        scanf("%d %d", &m, &n);
```

```
        if (m <= 0 || n <= 0 || m > n)
```

```
        {
```

```
            printf("Invalid input. Enter positive integers  
with m <= n.\n");
```

```
        }
```

```
    } while (m <= 0 || n <= 0 || m > n);
```

```
    /* Menu */
```

```
    do
```

```
    {
```

```
        printf("\n1. Multiplication tables of m and n\n");
```

```
        printf("2. Multiplication grid\n");
```

```
        printf("3. Consecutive integers\n");
```

```

printf("4. Exit\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        printf("\nMultiplication Tables of %d and %d\n\n", m, n, m, n);

        for (i = 1; i <= 10; i++)
        {
            printf("%d x %2d = %-5d\n", m, i, m * i, n, i, n * i);
        }
        break;

    case 2:
        printf("\nMultiplication Grid (m=%d, n=%d)\n\n", m, n);

        /* Column headings */
        printf(" ");

        for (j = 1; j <= n; j++)
        {
            printf("%4d", j);
        }

        printf("\n");

        for (j = 1; j <= n; j++)
        {
            printf("----");
        }

        printf("\n");

        /* Grid */
        for (i = 1; i <= m; i++)

```

```

        {
            printf("%3d |", i);

            for (j = 1; j <= n; j++)
            {
                printf("%4d", i * j);
            }

            printf("\n");
        }
        break;

    case 3:
        printf("\nConsecutive Integers from %d to %d\n",
            (m=%d, n=%d)\n\n",
            m, n, m, n);

        for (i = m; i <= n; i++)
        {
            printf("%d:", i);

            for (j = 0; j <= 5; j++)
            {
                printf(" %d", i + j);
            }

            printf("\n");
        }
        break;

    case 4:
        printf("\nExiting program.\n");
        break;

    default:
        printf("\nInvalid menu choice. "
            "Please enter a number from 1 to
4.\n");
    }

    } while (choice != 4);

    return 0;
}

```

5. Execution

Choice 1

```
[aaryanrajput@aaryans-macbook ~ % nano dayy22.c
[aaryanrajput@aaryans-macbook ~ % clang dayy22.c -o day22.c
[aaryanrajput@aaryans-macbook ~ % ./dayy22.c
zsh: permission denied: ./dayy22.c
[aaryanrajput@aaryans-macbook ~ % ./day22.c
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 3 and 5 (m=3, n=5)

3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50
```

Choice 2

```
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

Multiplication Grid (m=3, n=5)

      1  2  3  4  5
-----
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: █
```

Choice 3

```
Multiplication Grid (m=3, n=5)

      1  2  3  4  5
-----
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)

3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: █
```

Invalid Choice

```
[aaryanrajput@aaryans-macbook ~ % ./day22.c
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 9

Invalid menu choice. Please enter a number from 1 to 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: █
```

Choice 4

```
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10
```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 4

Exiting program.

```
[aaryanrajput@aaryans-macbook ~ % ./day22.c
```

```
Enter two positive integers m and n (m <= n): 3 5
```